

Paperclip plug-in

This plug-in adds an image to the event, and shows a thumbnail. It helps in handling file attachments and their storage on the server's file-system. It allows uploaded files to be handled like regular model fields/properties. It also includes methods to validate file size, file type, etc. Paperclip proves to be excellent while handling image attachments. It uses Imagemagick to resize images and generate thumbnails.

Installation

Install the plugin from the online git repository:

```
script/plugin install https://github.com/thoughtbot/paperclip.git
```

This will install the paperclip plugin under events/vendor/plugins

Usage

Let us add an image attachment to our *Event* model:

```
script/generate paperclip event image
```

Paperclip comes with a generator, and this command will create a migration named *TIMESTAMP_add_attachments_image_to_event.rb* in the *events/db/migrate/* directory. This migration will add the following fields to the *Event* model:

- image_file_name
- image_content_type
- image_file_size
- image_updated_at

Run the migration, as follows:

```
rake db:migrate
```

We need to specify in the *Event* model that it uses an image attachment. The code to be added to the model is:

```
# events/app/models/event.rb
has_attached_file :image,
:styles => {:thumbnail => "100x100#", :large => "500x500#"}
```

The *styles* parameter mentions that we need a thumbnail of size 100*100 pixels, and a large image of size 500*500 pixels. *'thumbnail'* and *'large'* are labels, and could be words of the user's choice. Let's add in a few validations:

```
# events/app/models/event.rb
validates_attachment_size :image, :less_than => 5.megabytes,
:message => "Image should be present and less than 5
MB in size"
validates_attachment_content_type :image,
:content_type => ['image/jpeg', 'image/png', 'image/
jpg'],
```

```
:message => "Attachment should be a jpg, jpeg or a
png"
```

The first validation will verify that the image size is less than 5 MB, and the second will verify that only JPEG and PNG images are stored.

Changes to views

To facilitate the upload of the image, we need to make the 'new event' and 'edit event' forms multi-part. Replace the following code...

```
# events/app/views/events/new.html.erb and events/app/views/
events/edit.html.erb
<% form_for(@event) do |f| %>
```

...with:

```
# events/app/views/events/new.html.erb and events/app/views/
events/edit.html.erb
<% form_for(@event, :html => { :multipart => true } ) do |f| %>
```

Then add the file field given below:

```
# events/app/views/events/new.html.erb and events/app/views/
events/edit.html.erb
<%= f.file_field :image %>
```

To display the image thumbnail in the show and index views, you should add the following helper:

```
# events/app/views/events/show.html.erb and events/app/views/
events/index.html.erb
<%= image_tag @event.image.url(:thumbnail) %>
```

To learn more, check <https://github.com/thoughtbot/paperclip/wiki>.

Will_Paginate plug-in

This plug-in will allow us to implement pagination, i.e., to break a long list of data items into numbered pages. We shall restrict listings to five events per page in the Event index view.

Installation

Install will_paginate from the online git repository:

```
script/plugin install git://github.com/mislav/will_paginate.git
```

This will install the will_paginate plugin under events/vendor/plugins

Usage

In the index method of *EventsController* (*events/app/controllers/events_controller.rb*), replace the following code...